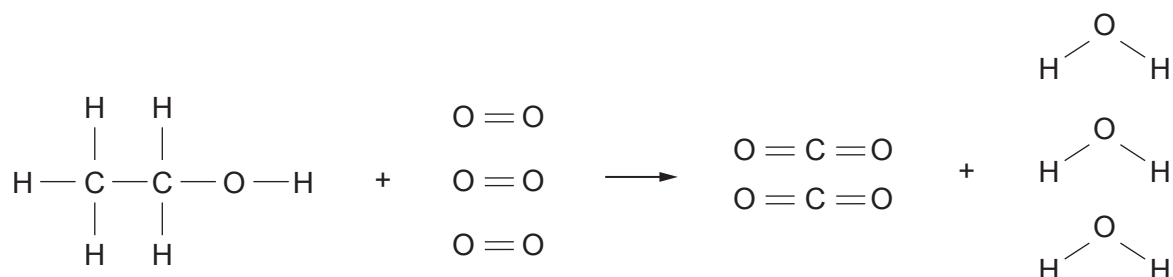


5. Ethanol is widely used in biofuels. When ethanol is burned in a car engine, carbon dioxide and water are formed.

The displayed formula equation for this reaction is shown below.



Some of the bond energies for this reaction are shown in the table below.

Bond	Bond energy (kJ/mol)
C–C	347
C–H	413
C=O	732
C–O	336
O–H	464

Use the information above to answer the following questions.

- (a) The energy needed to break all the bonds in the reactants is 4706 kJ. Calculate the energy needed to break an O=O bond. [4]

energy = kJ



(b) The energy released to form new bonds in the products is 5712 kJ/mol.

Explain why a car engine using ethanol as a fuel needs a cooling system.

[3]

Examiner
only

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